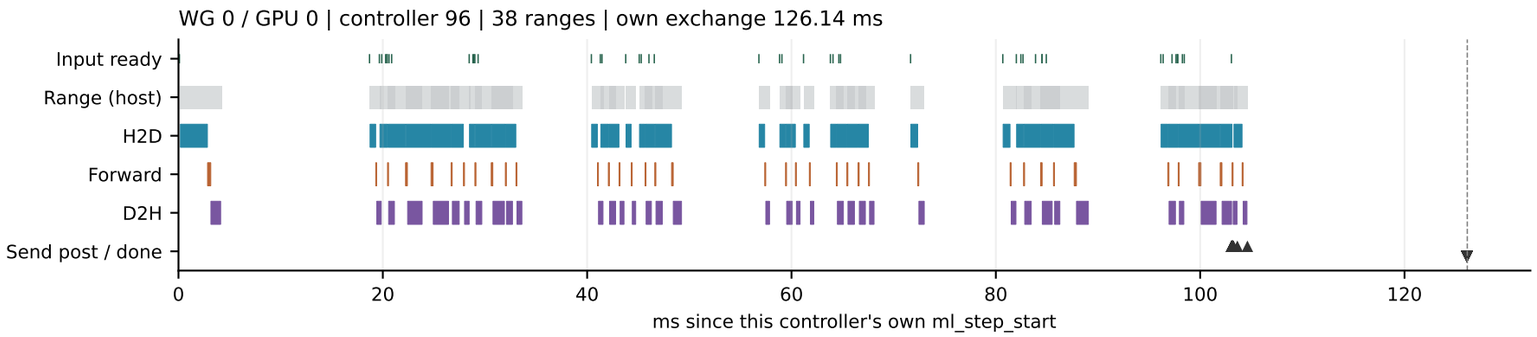
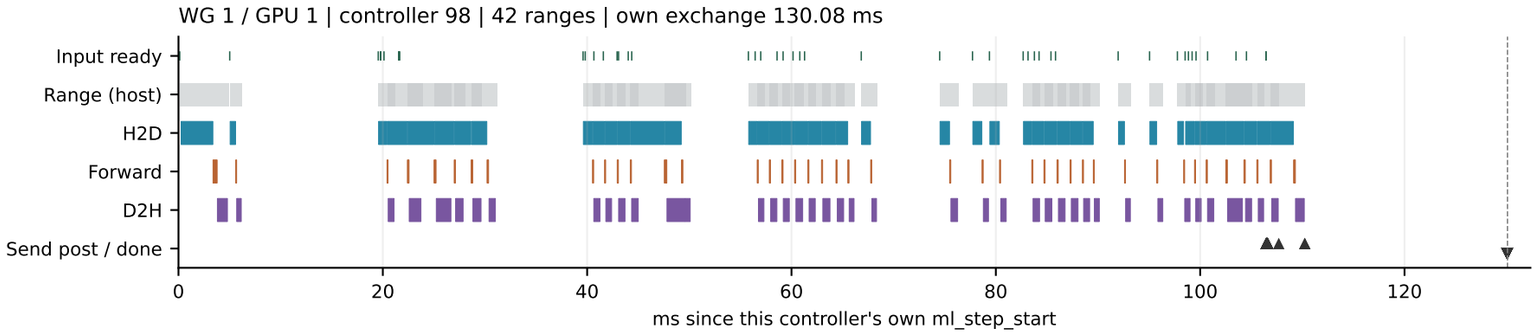
mlp | 96p96c4g | numa | collective vs p2pfull | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 264.76 ms | p2pfull 192.99 ms



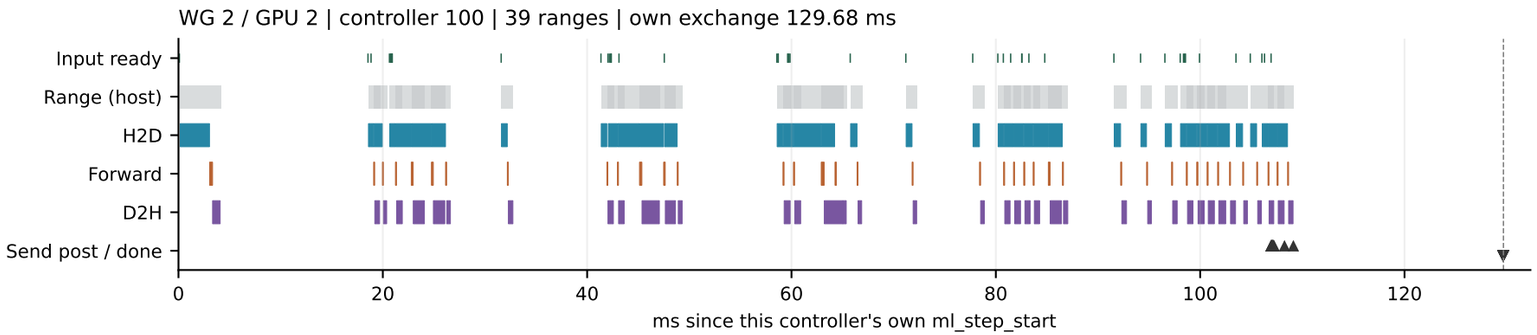
GPU work (ms)	Collective	p2pfull
H2D	26.546	42.023
Forward	2.606	4.713
D2H	13.454	24.783
GPU SUM	42.606	71.519

Controllers: coll 96 / P2P 96
Coll host device elapsed: 42.762 ms
P2P GPU envelope: 104.390 ms
Different scopes; not an elapsed-time comparison.



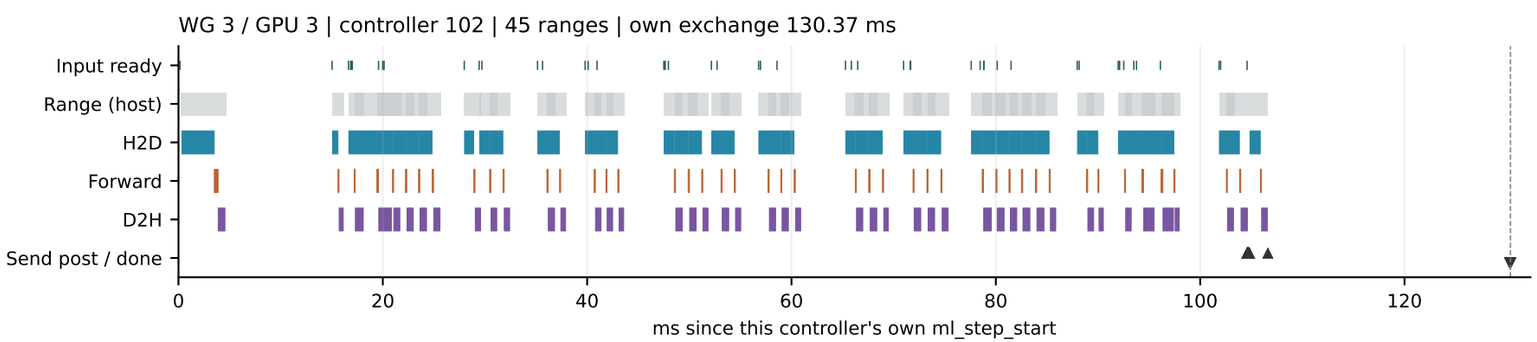
GPU work (ms)	Collective	p2pfull
H2D	25.584	54.725
Forward	2.595	5.462
D2H	13.109	28.800
GPU SUM	41.288	88.987

Controllers: coll 98 / P2P 98
Coll host device elapsed: 41.468 ms
P2P GPU envelope: 109.933 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	26.591	40.645
Forward	2.614	4.749
D2H	13.412	23.645
GPU SUM	42.618	69.039

Controllers: coll 100 / P2P 100
Coll host device elapsed: 42.806 ms
P2P GPU envelope: 108.920 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	25.855	53.162
Forward	2.592	5.467
D2H	12.905	27.778
GPU SUM	41.352	86.406

Controllers: coll 102 / P2P 102
Coll host device elapsed: 41.534 ms
P2P GPU envelope: 106.236 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.